

Havoc

Product Requirements Document

A live multiplayer party-game app where friends create private parties and compete in structured mini-games.

Product thesis. Havoc wins when every session feels like a tiny live game show: fast to understand, genuinely competitive, funny even in failure, and naturally worth sharing.

Document status: Founder-ready product definition; pricing and launch targets are planning hypotheses.

Launch scope: friends-first US closed beta on iOS and Android; invite-only Party Mode; 30 enabled games; 100-game documented catalog.

North star: Completed friend-party games per weekly active player.

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1. Executive summary

Havoc is a mobile-first multiplayer party-game app built around short, structured competitions over live video and voice. Friends create or join an invite-only party, jump into a rapid sequence of mini-games, react to the results, and run back the challenges that create the best group moments.

The friends-first beta launches with 30 polished games spanning face tracking, gesture and pose, object hunts, rhythm and voice, touch and motion, and opt-in GPS competitions inside private parties. A reusable Game Definition platform allows Havoc to ship two new games or meaningful variants every month without rebuilding party orchestration, scoring, safety, or analytics.

Havoc is fun-first, but fun depends on trust. Scoring must feel correct, party joining must be instant, group audio and video must stay stable, and private information must not leak. Safety, moderation, anti-cheat, and reliability therefore operate as invisible game infrastructure rather than as the product's emotional center.

1.1 Decisions locked for beta

- Positioning and sequencing: Havoc launches as invite-only Party Mode for friends. Random-person matchmaking is a later, separately gated expansion - not part of the MVP or initial beta.
- Platforms: simultaneous controlled iOS and Android beta.
- Age access: all launch play occurs in invite-only parties. Ages 13-15 use guardian-controlled approved contacts; older users invite friends under age-compatible party rules. Future public pools remain separate and require new launch approval.
- Catalog: 30 enabled launch games and 100 total documented concepts.
- Advertising: an interstitial opportunity after every two completed standard games, never during active competition.
- Subscriptions: No Ads at \$1.99/month or \$14.99/year; Havoc+ at \$4.99/month or \$34.99/year.
- GPS: private parties support opt-in distance and step competitions without exposing routes or coordinates to friends.
- Growth: party invitations and Havoc Highlights form the primary acquisition loop; every shared clip or challenge can deep-link into a friend party.
- Fairness: purchases never affect party placement, scoring, rank, game selection, or future matchmaking difficulty.

Core loop: Start Party -> invite friends -> ready lobby -> game reveal -> 3...2...1 -> live competition -> result and Highlight -> revenge vote or next game.

2. Vision, principles, and non-goals

2.1 Product vision

Make real-time social competition as easy to start as scrolling a feed and as memorable as playing party games in the same room. Havoc should become the default answer to: 'We are bored - what can we play right now?'

2.2 Fun design doctrine

- Five-second comprehension: a player can describe the goal after one sentence and one visual example.
- Three-second ignition: every round begins with the same unmistakable 3...2...1 ritual.
- Short dramatic arc: setup, tension, twist, resolution, and reaction occur within 20-90 seconds.
- Active cameras: the opponent's effort and reactions are part of the game, not decoration.
- Skill with sparkle: target roughly 70% controllable skill and 30% announced, symmetrical surprise.
- Funny failure: missing a pose or grabbing the wrong object should generate a better moment than a silent error.
- Readable fairness: timers, confidence, penalties, and No Contest decisions are visible and explainable.
- Immediate revenge: losing should create a reason to rematch, not a reason to leave.
- Freshness by construction: no random repeat inside the last ten matches unless a rematch overrides it.
- Highlight readiness: each game's design identifies at least one likely shareable beat.
- Zero pay-to-win: money buys expression and convenience, never outcomes.

2.3 Non-goals

- Public stranger matchmaking or an unlimited random video-chat room during the friends-first launch.
- A dating app, attractiveness-rating product, or gender-filter marketplace.
- A passive short-video feed that displaces the live game loop.
- A casino, wagering system, tradable economy, or randomized loot-box product.
- A location-discovery or nearby-people service.
- A game engine for arbitrary executable user code during beta.
- A claim of zero bugs or zero attacks; the product instead commits to measurable quality and incident response.

3. Audience and player motivations

3.1 Primary audiences

Persona	Age focus	Job to be done	Best entry points
Competitive Connector	16-20	Wants instant opponents, visible skill, ranks, and rematches.	Quick Chaos, Party Rivalry, Gauntlet
Friend-Group Instigator	13-22	Needs something everyone can join without explaining a complicated game.	Private Party, Squad Clash, Daily Havoc
Reaction Creator	16-24	Looks for authentic moments that convert into short-form content.	Highlights, Creator Challenge
Casual Chaos Player	16-24	Wants laughter and novelty more than mastery.	Wildcard playlists, object and comedy games
Skill Specialist	16-24	Returns to master a game family and defend a visible record.	Family rating, Rivalries, tournaments

3.2 Access model

- Ages 13-15: guardian-controlled account; approved contacts and private age-compatible parties only; public discovery disabled.
- Ages 16-17: invite-only friend parties at launch with age assurance proportional to party and regional requirements; no stranger discovery.
- Ages 18+: invite-only friend parties at launch; future public matchmaking, if approved, uses a separate adult pool with no teen visibility.
- Exact birth date, school, precise location, and legal identity are hidden from other players.
- Public profiles show a pseudonymous handle, broad age band, language, badges, sportsmanship record, and optional interests.
- Age-mismatch signals trigger re-verification, temporary party-access restriction, or manual review.

3.3 Jobs to be done

- When I have two spare minutes, give me a real opponent and a challenge immediately.
- When my group call gets stale, turn it into a party game without making everyone download rules.
- When I lose narrowly, let me prove it was a fluke.
- When something hilarious happens, help me share it without editing a video myself.
- When I care about a game, let me improve and see a meaningful record.
- When the network or detector is uncertain, protect the legitimacy of the result.

4. Competitive research and product lessons

The launch opportunity is not to rebuild Omegle or solve stranger-marketplace liquidity. It is to combine the spontaneous friend presence of Houseparty, the content breadth of Hago, and the shareability of modern social products - while making the game itself the reason a friend group assembles and returns. Random-person play remains a later option only after separate policy, safety, liquidity, and trust gates.

Product	What worked	Failure mode	Havoc decision
Omegle	Instant novelty, no learning curve, creator reaction content.	Underfunded moderation, anonymity, and weak monetization made abuse and operating cost existential.	Keep instant matching; reject open-ended anonymity.
Monkey	Timed encounters, mutual extension, youth-native visual language, off-platform sharing.	Short timers did not prevent sexual content; social-handle movement weakened platform control.	Use timed games and mutual actions; keep strong verification and kill switches.
Houseparty	Spontaneous friend presence, group video, embedded games, low-friction joining.	Reliance on social presence and external conditions limited durability.	Make friend parties a retention engine, not a secondary menu.
Yubo	Age assurance, AI and human moderation, subscriptions, virtual currency, boosts.	Paid visibility and social-discovery framing can distort incentives.	Borrow verification and commerce; avoid paid matchmaking advantage.
Hago	Large game catalog, voice rooms, group play, ads and purchases.	Frequent ads can become the dominant user complaint.	Maintain breadth, but instrument the two-game ad cadence as a guardrail experiment.
OmeTV	Fast matching and simple value proposition; premium visibility products.	Random-chat framing and mature-content risk constrain distribution.	Position as structured competition and avoid unrestricted chat.

Policy reality. Apple states that Chatroulette-style, random, or anonymous chat experiences may not belong on the App Store. [Apple App Review Guidelines](#) and [Apple's random-chat clarification](#) reinforce why Havoc must be presented and operated as a game product.

Child-safety expectations. Google requires random-chat and social products to publish child-safety standards and operate reporting and response systems. See [Google Play Child Safety Standards](#) and [Google Play UGC policy](#).

The Omegle lesson. Australia's eSafety Commissioner reported that Omegle had only a handful of moderators despite tens of thousands of concurrent users. [eSafety's Omegle shutdown statement](#) makes moderation staffing a core cost assumption.

5. Core experience and competitive modes

5.1 Match lifecycle

1. Party entry: start a party or open an invite link, then choose Quick Chaos, Best of Three, GPS, or a featured playlist.
2. Party lobby and preflight: show who has joined, run an interactive warmup, verify camera framing, microphone, permissions, environment, and readiness, then let the host start.
3. Reveal: show the game name, one-line rule, visual example, duration, and any announced Wildcard modifier.
4. Countdown: synchronize both clients to a server-authoritative 3...2...1 with audio, haptic, and visual signals.
5. Live round: transmit media and signed game events; show progress without obscuring opponent reactions.
6. Verification: reconcile client observations, server time, confidence, network conditions, and anti-cheat signals.
7. Result: deliver a dramatic win, loss, draw, or No Contest with understandable cause.
8. Aftermath: reactions, Highlight approval, revenge vote, next party game, team reshuffle, or a clean exit.
9. Ad opportunity: after every second completed standard game, outside decisive sets and safety flows.

5.2 Mode requirements

Mode	Format	Primary fun job	Target block
Quick Chaos	1v1 or rotating pair inside a party; one random eligible game	Fastest party start; default lobby CTA	20-90 sec
Best of Three	Same rival, three distinct families	Narrative and comeback potential	3-6 min
Revenge Match	Loser chooses one of three offers	Turns loss into agency	30-90 sec
Havoc Gauntlet	Five games, cumulative points	Session anchor and daily competition	7-10 min
Party Rivalry	Friend head-to-head records and family-specific ratings	Mastery and durable records	1-5 games
Wildcard Round	Symmetrical announced modifier	Novelty without hidden unfairness	Varies
Duos	2v2 coordinated or relay scoring	Friend acquisition and teamwork	2-5 min
Squad Clash	Teams of 3-4 in private parties	Group retention	5-15 min

Mode	Format	Primary fun job	Target block
Elimination Party	Lowest score exits active slot but gains spectator role	Drama without dead spectators	8-15 min
King of the Party	Winner stays; challengers rotate	Fast party leadership loop	Open-ended blocks
Tournament	8- or 16-player bracket	Scheduled friend, creator, and community-party events	15-30 min
Daily Havoc	Shared five-game seed	Comparable friend scores and habit	5-8 min
Creator Challenge	Curated branded game pack	Acquisition and sponsorship	3-10 min
Practice Lab	Solo calibration and drills	Accessibility, onboarding, and mastery	Optional

5.3 Party formation and room assignment

- Party creation: generate an expiring invite link and short code, expose the native share sheet, and show a challenge preview before account creation.
- Joining: resolve the invite into the correct private room, minimize onboarding steps, and preserve the player's place while required permissions are granted.
- Room assignment never uses physical proximity; region is used only for latency, legal policy, and media routing.
- Do not sell host power, easier scoring, priority turns, safety overrides, or teen visibility.
- When a party has only two or three players, automatically select compatible 1v1, co-op, and rotating-seat formats instead of making the room feel incomplete.
- While friends join, use playable warmups, predictions, Daily Havoc previews, and host controls rather than a passive waiting screen.

6. Complete feature requirements

6.1 Onboarding and calibration

- Create a pseudonymous account, collect birth date through a neutral screen, and complete age estimation plus liveness.
- Use ID verification only as a fallback; retain verification artifacts according to vendor and legal policy rather than as profile media.
- Teach camera framing, lighting, microphone, safe movement, reporting, party invitations, host controls, and the distinction between launch Party Mode and any future public mode.
- Run a 30-second Practice Lab sampler that calibrates blink, pose, hands, microphone, latency, and device motion.
- Ask for permissions in context immediately before the first game that needs them; do not require GPS at registration.

- Let players mark accessibility and environment preferences, such as seated play, no audio, limited space, or reduced motion.

6.2 Home and discovery

- Primary CTAs: Start Party and Join Party.
- Secondary modules: friends online, recent parties, Daily Havoc, featured playlist, current season, and latest game drop.
- Use game trailers, predictions, and interactive warmups while friends join instead of passive loading spinners.
- Show progress by game family, not a cluttered global inventory.
- Expose explicit 'more like this' and 'less like this' controls after games to improve variety without opaque engagement profiling.

6.3 Friends, rivals, and social continuity

- Players select existing friends or share an invite before play; recent party members can be re-invited with one tap, while block controls remain immediately available.
- Verified users aged 16+ may mutually reveal supported social handles after a warning; store the reveal event and maintain a remote kill switch.
- Structured in-app reactions are available immediately; free-form messaging is filtered, rate-limited, and disabled during active games.
- Rivalries require mutual acceptance and display head-to-head record, favorite game families, and rematch CTA.
- Party links and codes expire, are revocable by the host, are rate-limited against spam, and respect age-tier and guardian constraints.

6.4 Progression and rewards

- Separate ratings for Reflex, Performance, Scavenger, Movement, Sound, and Party skill families.
- Player level reflects participation and sportsmanship, while rank reflects competitive performance.
- Season tracks grant known rewards at visible milestones; no randomized loot boxes.
- Achievements reward variety, close wins, respectful play, comeback wins, party hosting, and Highlight creativity.
- Daily Havoc offers a shared challenge, but missing a day never destroys a streak or removes earned value.

6.5 Spectator play

- Eliminated party members remain active through winner predictions, reaction bursts, and harmless pre-approved modifiers.

- Spectator actions cannot obscure safety controls, reveal answers, or materially favor a team unless the mode explicitly grants symmetric power.
- Large party and tournament spectators receive role-limited interaction; hosts may enable richer pre-approved reactions and predictions.

7. UX, visual identity, motion, and sound

7.1 Controlled-chaos design system

- Foundation: near-black gameplay canvas so live video and bright states dominate.
- Accents: electric lime for go/success, ultraviolet for identity, cobalt for information, and hot coral for dramatic warnings or loss.
- Typography: oversized condensed countdown and score moments; highly legible neutral UI type elsewhere.
- Motion: sharp anticipation, elastic result impacts, freeze frames, confetti, score punches, and instant replay wipes.
- Sound: short recognizable game identities, crowd swells, fakeout cues, and an announcer voice used sparingly.
- Haptics: consistent vocabulary for countdown, valid action, penalty, close finish, win, and safety stop.

7.2 Screen map

Screen	Core content	Emotional job
Welcome and age assurance	Identity, age tier, consent, and community rules	Trust without administrative drag
Calibration Lab	Camera, audio, gesture, lighting, network, permissions	Turns setup into a first mini-game
Home	Start Party, Join Party, friends online, Daily, season	One obvious play button
Party lobby	Invite status, joined friends, warmup, ready state	Anticipation without dead time
Reveal	Game title, five-second rule, Wildcard	Anticipation
Live game	Video, timer, score, progress, leave/report	Action and reactions
Result	Winner, evidence, reactions, revenge	Drama and clarity
Highlight Studio	Clip options, identity controls, dual consent	One-tap shareability
Party lobby	Players, teams, playlist, host controls	Instant group coordination
Shop and pass	Known items, prices, preview	Expression without pay-to-win
Safety Center	Reports, blocks, appeal, controls	Fast recovery and transparency

7.3 Accessibility

- Captions for announcer prompts and player speech where enabled.
- Reduced-motion mode that preserves timing information without large kinetic effects.
- Color-blind-safe icons, labels, and patterns in addition to color.
- Text scaling without covering video, timer, or safety controls.
- Seated, low-space, silent, no-camera, and no-GPS playlist filters where compatible.
- Detector calibration that reports why a body part, face, or sound is not being recognized.
- No rank penalty when the system cancels an inaccessible or technically invalid game.

8. Thirty beta launch games

Every launch game below is a product, ML, fairness, and showmanship unit. A game is enabled in the friends-first beta only after its scoring model, anti-cheat policy, accessibility path, failure states, and fun scorecard pass the launch gate.

8.1 Stare Royale

Face / endurance | 20-60 seconds | 1v1, Gauntlet, tournament

Play: Both players center their face and stare into the camera. The first confirmed natural blink loses; if both blink within the fairness window, the round continues in sudden death.

Scoring: Binary win based on the first high-confidence blink timestamp. Server time reconciles both clients with a 180 ms simultaneity window.

Detection: Face presence, eye landmarks, eye-aspect ratio over multiple frames, head pose, occlusion, liveness, and capture quality.

Anti-cheat: Reject prerecorded loops, covered eyes, sunglasses, off-axis faces, low frame rate, and intentional camera blur. Random micro-liveness prompts appear before the countdown, never mid-stare.

Showmanship and Highlight: Heartbeat audio ramps over time; opponent blink triggers a freeze-frame zoom and 'CAUGHT BLINKING' stamp. Best Highlight is the loser's reaction.

Failure and No Contest: No Contest if either face is absent or eye confidence stays below threshold for more than 800 ms. Pause once for a network interruption; then cancel.

Safety and accessibility: Seated-friendly and low-motion. Offer a reduced-flash presentation and a no-audio version.

Analytics and launch gate: False-blink audits, simultaneous-blink rate, average hold time, rematch rate, and fun rating. Launch: audited false winner below 0.5%.

8.2 Don't Laugh

Face / comedy | 30-60 seconds | 1v1, party elimination, creator pack

Play: Both players keep a straight face while Havoc plays safe visual prompts, sound stings, and opponent-triggered reaction cards. First confirmed laugh or broad smile loses.

Scoring: Expression intensity accumulates; a sustained threshold or vocal laugh event ends the round. Near-threshold events add visible pressure but do not instantly decide it.

Detection: Face landmarks, mouth shape, cheek lift, eye squint, audio laugh classifier, baseline neutral calibration, and liveness.

Anti-cheat: Require face visibility and neutral baseline; block masks, still images, muted microphone when the playlist requires audio, and deliberate face cropping.

Showmanship and Highlight: Comedic prompts accelerate and the first break becomes a slow-motion replay. Party mode lets eliminated players submit approved reaction cards.

Failure and No Contest: If expression confidence is ambiguous, continue rather than penalize. Draw if both break within 250 ms.

Safety and accessibility: Prompt library excludes sexual, hateful, body-shaming, frightening, or culturally targeted humor. Caption every audio prompt.

Analytics and launch gate: Break rate by prompt, prompt fatigue, false-positive smile disputes, clip approvals, and repeat-play score. Launch: 4.4/5 fun target.

8.3 Emoji Face-Off

Face / accuracy | 30-45 seconds | 1v1, Best of Three, party

Play: A sequence of exaggerated expression cards appears. Players recreate each face; first to hit the similarity target earns the point. Best of five wins.

Scoring: Weighted landmark similarity, head orientation, timing, and held confirmation. Expressions are scored relative to each player's calibrated baseline.

Detection: Face landmarks, head pose, mouth/eye/brow geometry, capture quality, and liveness.

Anti-cheat: Prevent screen-image substitution and require natural motion into each pose. The target sequence is server seeded and unknown before reveal.

Showmanship and Highlight: Correct matches snap into a comic-book frame; misses receive playful labels such as 'too suspicious' or 'not shocked enough.'

Failure and No Contest: Skip expressions that either player cannot calibrate reliably; No Contest if fewer than three valid cards remain.

Safety and accessibility: Do not score beauty, attractiveness, race, gender presentation, or facial difference. Provide a gesture-based alternative playlist.

Analytics and launch gate: Similarity distribution, per-expression accessibility, time-to-match, disagreement reports, and Highlight rate.

8.4 Meme Face Race

Face / speed | 25-40 seconds | 1v1, creator challenge

Play: An absurd illustrated reaction appears for a fraction of a second. Players race to reproduce it from memory. Fastest valid match takes the card.

Scoring: Three cards; speed breaks ties after minimum similarity. A perfect-but-slow face can lose to a fast valid interpretation.

Detection: Landmark geometry, head angle, expression family, gaze direction, and baseline normalization.

Anti-cheat: Targets are generated server-side; require continuous live motion and reject imported camera streams.

Showmanship and Highlight: Side-by-side target and player freeze frame becomes the default Highlight, with a similarity meter and instant caption.

Failure and No Contest: If neither player reaches threshold within eight seconds, reveal the funniest near-match and award no point.

Safety and accessibility: Use licensed or original artwork, never real-person ridicule. Support reduced visual complexity.

Analytics and launch gate: Recognition time, valid-match rate, funniest-near-match shares, content fatigue, and per-card fairness.

8.5 Pose Panic

Body / accuracy | 30-60 seconds | 1v1, squads, elimination

Play: Players copy increasingly ridiculous full-body silhouettes before the timer closes. Each round adds a limb or orientation constraint.

Scoring: Joint-angle similarity plus time held. Best cumulative score across three poses wins.

Detection: 2D/3D body pose where supported, person segmentation, joint confidence, orientation, and floor/space calibration.

Anti-cheat: Require live transitions, one primary person, and plausible joint motion. Reject static photos and multiple-person assistance.

Showmanship and Highlight: Skeleton trails reveal how each player moved into position; the final impossible pose gets a dramatic split-screen freeze.

Failure and No Contest: Offer upper-body substitutions when full-body framing fails. No Contest if both players lack usable space.

Safety and accessibility: No inversions, jumps near furniture, painful flexibility, or floor drops. Seated pose set available.

Analytics and launch gate: Pose completion, calibration exits, joint-confidence gaps, accessibility substitutions, and injury/safety feedback.

8.6 Mirror Me

Body / memory | 30-50 seconds | 1v1, Duos, party relay

Play: One player receives a short gesture sequence; the opponent must mirror it exactly. Roles swap, and the cleaner/faster mirror wins.

Scoring: Sequence order, mirrored joint path, timing error, and final pose hold produce a normalized score.

Detection: Upper-body pose, hand position, motion trajectory, and orientation.

Anti-cheat: Role-specific private prompts, continuous liveness, and server-timed sequence windows prevent preplanning.

Showmanship and Highlight: A motion trail overlays both players; the best synchronized moment becomes the Highlight.

Failure and No Contest: Replay the prompt once if delivery drops before the role begins; never replay after movement starts.

Safety and accessibility: Upper-body by default; avoid gestures that can be hateful, sexual, threatening, or culturally offensive.

Analytics and launch gate: Mirror error, role advantage, sequence difficulty, sync clip rate, and rematch conversion.

8.7 Freeze!

Body / control | 25-45 seconds | 1v1, party elimination

Play: Players move while the beat runs and must stop instantly on the freeze cue. Fake cues and visual distractions create pressure.

Scoring: Reaction time to the true cue plus movement after the stop line. Lowest valid motion wins each heat.

Detection: Body optical flow, pose velocity, audio cue clock, device motion, and frame timing.

Anti-cheat: Require visible movement before the cue and randomize the freeze point. Static players receive no score.

Showmanship and Highlight: Motion trails slam to a stop; late limbs receive comic arrows and a slow-motion replay.

Failure and No Contest: No score when pre-cue movement is below minimum or the camera frame is unstable.

Safety and accessibility: No jumping required. Offer seated upper-body movement and reduced-audio cueing.

Analytics and launch gate: Pre-cue participation, stop latency, false motion flags, eliminated-player retention, and fun score.

8.8 RPS Blitz

Hand / speed | 20-35 seconds | 1v1, Best of Three, tournament

Play: Rapid best-of-five rock-paper-scissors. Throws lock on the final beat; later rounds shorten the cadence.

Scoring: Standard RPS outcome with server-authoritative lock time. First to three points wins.

Detection: Hand landmarks, gesture classification, chirality, frame timing, and liveness.

Anti-cheat: Simultaneous hidden capture prevents reaction cheating; reject ambiguous or late hands.

Showmanship and Highlight: Hands collide as stylized icons; match point uses a dramatic pause before reveal.

Failure and No Contest: Ambiguous throws replay without a point. Repeated invalid hands trigger calibration or forfeiture.

Safety and accessibility: One-hand alternative and mirrored-camera explanation. No sensitive gesture mappings.

Analytics and launch gate: Invalid throw rate, side latency, draw rate, match-point exits, and replay rate.

8.9 Hand Signal Sprint

Hand / reaction | 30-45 seconds | 1v1, Gauntlet

Play: Prompts demand finger counts, thumbs, peace signs, or left/right combinations. Correct signal first earns the point.

Scoring: Five to seven prompts; accuracy precedes reaction time. Wrong confident gesture creates a short lockout.

Detection: Hand landmarks, finger states, chirality, two-hand presence, and timestamp.

Anti-cheat: Server-randomized prompts and continuous motion prevent printed or prerecorded answers.

Showmanship and Highlight: Correct hands glow; wrong signals burst into playful 'too many fingers' effects.

Failure and No Contest: Skip gestures that the player's calibration cannot recognize. Draw if fewer than four shared gestures remain.

Safety and accessibility: Gesture library is culturally reviewed and excludes offensive signs; voice prompt alternative available.

Analytics and launch gate: Recognition by device/skin tone/lighting, wrong-lock frequency, reaction distribution, and accessibility skips.

8.10 Simon Says Chaos

Body / inhibition | 35-60 seconds | 1v1, party elimination

Play: Perform commands only when preceded by 'Havoc says.' The announcer accelerates, changes voice, and inserts believable traps.

Scoring: Correct actions add points; trap actions subtract points. Fast valid completion breaks ties.

Detection: Pose and gesture events, ASR prompt clock, motion onset, and sequence state.

Anti-cheat: Prompt order is server seeded and action windows are short; require visible readiness and live transitions.

Showmanship and Highlight: Trap failures trigger an instant buzzer, red outline, and reaction replay.

Failure and No Contest: If the commanded gesture is not reliable for either player, swap it before the round.

Safety and accessibility: Commands remain low-impact and space-aware. Never ask players to leave the camera, touch others, or use unsafe objects.

Analytics and launch gate: Trap success, prompt comprehension, ASR delivery latency, elimination timing, and laugh/Highlight rate.

8.11 Dance Freeze

Body / performance | 40-75 seconds | 1v1, squads, creator pack

Play: Dance while music plays, hit prompted move moments, and freeze cleanly on the cutoff. Style is not judged; participation, timing, and freeze control are.

Scoring: Movement participation, beat alignment, prompt completion, and post-cutoff motion create a fair technical score.

Detection: Pose velocity, beat clock, motion energy, prompt gestures, and freeze response.

Anti-cheat: Require live motion and prevent prerecorded media input. Static players cannot win by gaming the freeze score.

Showmanship and Highlight: The best move and worst freeze become alternate Highlights with beat-synced cuts.

Failure and No Contest: Audio desync above threshold pauses and restarts the heat. Offer haptic/visual beat mode.

Safety and accessibility: Low-impact movement only; seated groove mode and no music lyrics requiring licensing.

Analytics and launch gate: Participation energy, beat error, freeze accuracy, seated-mode usage, and clip share rate.

8.12 Copycat Relay

Body / sequence | 45-75 seconds | 1v1, Duos, squads

Play: Havoc starts a gesture chain. Each player repeats it and adds one approved move; the sequence grows until someone misses.

Scoring: Longest fully correct chain wins; timing and pose accuracy resolve equal rounds.

Detection: Gesture sequence, body/hand pose, role ownership, and motion order.

Anti-cheat: Limit move vocabulary to detected safe gestures; hide the next player's camera overlay during recall if required.

Showmanship and Highlight: The full chain replays at high speed, ending on the exact miss.

Failure and No Contest: If a user-created move is not confidently recognized, Havoc substitutes a detected template before continuing.

Safety and accessibility: No free-form gestures that are dangerous or offensive. Seated vocabulary supported.

Analytics and launch gate: Average chain length, recognition substitutions, role fairness, party retention, and replay shares.

8.13 Find It Fast

Object / scavenger | 30-60 seconds | 1v1, squads, Daily Havoc

Play: A common-object target appears. First player to retrieve it, return on camera, and hold it inside the scan frame wins.

Scoring: First high-confidence correct object after target reveal; server timestamp breaks close finishes.

Detection: Curated object detector, object size/position, hand presence, OCR where appropriate, and liveness.

Anti-cheat: Targets are revealed after countdown; require object movement into frame and occasional angle prompt. Reject screen photos.

Showmanship and Highlight: Correct object gets a game-show scan beam; close finishes use a split-screen photo finish.

Failure and No Contest: If the detector cannot confirm either object, offer human self-report only as feedback, not as a ranked result.

Safety and accessibility: Target list excludes medicine, weapons, addresses, and family photos.

Analytics and launch gate: Target availability, detector precision, room-leave time, close finishes, and target opt-outs.

8.14 Color Dash

Object / speed | 20-40 seconds | 1v1, party relay

Play: Find and show an object dominated by the target color. Multi-round mode uses increasingly specific shades.

Scoring: Color coverage, color distance, object presence, and valid return time.

Detection: Foreground object segmentation, color-space distance, hand/object motion, and camera white-balance calibration.

Anti-cheat: Reject full-screen color displays, camera covers, and static pre-positioned swatches; require a real object boundary and movement.

Showmanship and Highlight: The winning object fills the UI with its color; near misses receive a humorous shade name.

Failure and No Contest: Use broad color families when either camera has unstable white balance.

Safety and accessibility: Players may skip unavailable colors without penalty in private play; color names always include a visual pattern/icon.

Analytics and launch gate: Color bias by device/lighting, screen-cheat attempts, target availability, and relay pace.

8.15 Scavenger Stack

Object / sequence | 60-90 seconds | 1v1, Duos, squads

Play: Retrieve three items in the revealed order and stack or arrange them inside the scan zone.

Scoring: One point per verified item plus sequence and completion-time bonuses. Incorrect order must be repaired before finish.

Detection: Multi-object tracking, object classes, spatial order, hand motion, and scene continuity.

Anti-cheat: Random target sets, live item entry, and scene continuity prevent prepared still images.

Showmanship and Highlight: Each verified item locks into an on-screen slot; the final stack receives a tower animation and photo finish.

Failure and No Contest: If one target is unavailable, both players receive the same substitute. No ranked score if substitution parity cannot be guaranteed.

Analytics and launch gate: Completion rate, substitution rate, detector confusion, average travel time, and team coordination.

8.16 Texture Hunt

Object / perception | 30-50 seconds | 1v1, party

Play: Find something fuzzy, shiny, striped, transparent, rough, or soft and present it for verification.

Scoring: First item meeting the visual texture classifier and object-boundary requirements wins.

Detection: Material/texture classifier, segmentation, specularities, pattern features, and hand motion.

Anti-cheat: Reject screen images and flat printed textures; require angle changes or a gentle touch prompt.

Showmanship and Highlight: The UI magnifies the winning texture and adds an exaggerated tactile sound effect.

Failure and No Contest: Some textures remain unranked until classifier precision is proven; use peer voting only in private creative mode.

Safety and accessibility: No sharp, hot, dirty, chemical, or unknown substances. Provide a seated nearby-object playlist.

Analytics and launch gate: Texture precision, target availability, peer/model disagreement, and object-safety reports.

8.17 Alphabet Hunt

Object / knowledge | 30-60 seconds | 1v1, Daily Havoc

Play: A letter appears. Find an object whose common name starts with that letter and say its name while showing it.

Scoring: Object verification plus speech match; fastest valid pair wins. Regional synonyms are supported by locale.

Detection: Object detection, speech recognition, pronunciation variants, and language model constrained to safe object names.

Anti-cheat: Object and spoken label must agree; reject screenshots and text-only answers.

Showmanship and Highlight: The letter transforms into the recognized object name; creative valid synonyms become shareable moments.

Failure and No Contest: If both objects are plausible but unsupported, issue No Contest and add examples to the review queue.

Safety and accessibility: Block personal documents, medicine, weapons, and intimate objects. Accessibility mode permits pointing to nearby objects.

Analytics and launch gate: Unsupported synonym rate, language fairness, object-label agreement, and educational novelty.

8.18 Object Bingo

Object / strategy | 60-90 seconds | 1v1, Duos, squads

Play: A compact 3x3 board contains safe object or property targets. Complete any line before the opponent.

Scoring: First verified line wins; full-board progress resolves simultaneous lines.

Detection: Object, color, shape, and texture pipelines with persistent board state.

Anti-cheat: Targets are assigned after match; require live object entry and unique physical instances where specified.

Showmanship and Highlight: Cells stamp in with sounds, and the winning line burns across the board over the live reactions.

Failure and No Contest: Replace unavailable cells symmetrically once per match. Ranked play requires a validated target availability score.

Safety and accessibility: Board generator excludes risky targets and considers the player's declared environment and mobility.

Analytics and launch gate: Board completion distribution, unavailable cells, route strategy, and squad communication.

8.19 Clap Attack

Audio / rhythm | 30-50 seconds | 1v1, Best of Three, party

Play: Repeat and extend increasingly difficult clap patterns. A missed beat or wrong count loses the heat.

Scoring: Onset timing, pattern order, and amplitude consistency; longest valid sequence wins.

Detection: Audio onset, clap classifier, tempo alignment, echo cancellation, and microphone quality.

Anti-cheat: Patterns are server generated; distinguish speaker playback from near-field claps using spectral and timing features.

Showmanship and Highlight: Each clap creates a bright visual pulse; the exact missed beat explodes out of the rhythm lane.

Failure and No Contest: If the environment is too noisy, switch to screen taps with the same pattern.

Safety and accessibility: Volume warning and sensory-friendly visual-only mode.

Analytics and launch gate: Onset accuracy, noisy-room switches, playback-cheat flags, sequence length, and party replay rate.

8.20 Tongue-Twister Turbo

Voice / accuracy | 25-45 seconds | 1v1, creator pack

Play: Read a short original tongue twister quickly and clearly. The opponent receives the same text and language variant.

Scoring: Word accuracy first, then completion time; repeated or skipped words add transparent penalties.

Detection: On-device or streaming ASR, forced alignment, voice activity, pronunciation confidence, and noise quality.

Anti-cheat: Prompt appears only at reveal; detect synthetic playback and require visible live mouth movement when camera is enabled.

Showmanship and Highlight: Words light up in real time and fumbled syllables tumble off the screen.

Failure and No Contest: Do not rank when ASR confidence is below language/accent threshold. Offer tap-rhythm alternative.

Safety and accessibility: Original phrases avoid slurs, innuendo, stereotypes, and copyrighted lyrics. Captions are intrinsic.

Analytics and launch gate: ASR gap by accent, error distribution, phrase difficulty, replay rate, and dispute rate.

8.21 Hum Match

Voice / pitch | 30-45 seconds | 1v1, Duos

Play: Hear a short original melody, then hum it back. Players can steal a point if the opponent misses.

Scoring: Pitch contour, interval accuracy, rhythm, and completion. Transposition is allowed so vocal range does not decide the winner.

Detection: Fundamental pitch, onset, contour matching, noise suppression, and voice activity.

Anti-cheat: Detect direct speaker playback and require a brief live vocal warmup before the round.

Showmanship and Highlight: Pitch trails race along a neon path; a perfect contour triggers a crowd swell.

Failure and No Contest: If pitch cannot be extracted, switch to Beat Repeat without rank loss.

Safety and accessibility: No requirement for loud volume or extreme range. Visual and tap alternatives available.

Analytics and launch gate: Contour accuracy, transposition usage, microphone failures, replay detection, and duet Highlight rate.

8.22 Beat Repeat

Audio / memory | 30-55 seconds | 1v1, party relay

Play: Repeat a rhythm through taps, claps, or vocal sounds. Each successful round adds another beat.

Scoring: Sequence correctness, tempo tolerance, and longest chain.

Detection: Unified onset stream across microphone and touch inputs, tempo normalization, and event order.

Anti-cheat: Random server patterns, playback detection for audio mode, and no visible pattern after performance begins.

Showmanship and Highlight: Inputs animate a shared rhythm track; the miss freezes at the exact beat.

Failure and No Contest: Players select an accessible input method before matching; compare normalized timing, not raw sensor latency.

Safety and accessibility: Sensory mode lowers sound and replaces flashing beats with stable shapes.

Analytics and launch gate: Input-method fairness, sequence length, latency normalization, and group relay completion.

8.23 Sound Hunt

Audio / scavenger | 30-60 seconds | 1v1, party

Play: Produce or locate an everyday sound such as keys jingling, paper crumpling, water pouring, or a door knock.

Scoring: First high-confidence target sound after reveal wins; volume does not improve score.

Detection: Curated audio-event classifier, near-field quality, duration, and replay fingerprints.

Anti-cheat: Random target reveal, speaker-playback detection, and occasional camera confirmation for object-linked sounds.

Showmanship and Highlight: A waveform transforms into the recognized object; unusual authentic sounds make strong Highlights.

Failure and No Contest: If the classifier is uncertain, continue to a substitute sound rather than use peer judgment in ranked play.

Safety and accessibility: Exclude alarms, glass breaking, screaming, animals, appliances, dangerous tools, and disruptive public behavior.

Analytics and launch gate: Classifier precision, environment noise, substitute rate, target safety feedback, and shareability.

8.24 Say It Backwards

Voice / wordplay | 30-50 seconds | 1v1, Gauntlet

Play: Reverse short words or syllable groups under time pressure. Difficulty rises from spelling reversal to spoken phoneme reversal.

Scoring: Correct sequence first; ASR confidence and completion time determine points.

Detection: Constrained ASR, phoneme alignment, voice activity, and prompt timing.

Anti-cheat: Prompts are server generated and expire quickly; detect playback and background assistant audio.

Showmanship and Highlight: Correct answers visually rewind into the original word; wrong answers get a playful alternate transcription.

Failure and No Contest: Spelling-tap alternative for speech-incompatible environments; do not compare incompatible input modes in ranked play.

Safety and accessibility: Safe vocabulary only, locale-specific pronunciation, and no literacy shaming.

Analytics and launch gate: Accuracy by difficulty/language, alternate-input usage, completion time, and comedic transcription shares.

8.25 Reaction Flash

Touch / reflex | 20-35 seconds | 1v1, tournament

Play: Tap only when the true visual, haptic, or audio target appears. Fake cues punish anticipation.

Scoring: Median valid reaction time across five cues minus false-start penalties; server offsets device latency.

Detection: Touch timestamp, render acknowledgment, audio/haptic schedule, device refresh rate, and network clock.

Anti-cheat: Cryptographically seeded cue timing, impossible-response detection, automation heuristics, and integrity attestation.

Showmanship and Highlight: A photo-finish timeline displays both taps to the millisecond without overstating precision.

Failure and No Contest: No Contest when clock sync, frame pacing, or input telemetry fails tolerance.

Safety and accessibility: Visual, audio, and haptic cue sets; avoid strobing and startling volume.

Analytics and launch gate: Clock offset, false starts, device-class fairness, impossible inputs, and tournament confidence.

8.26 Tap Trap

Touch / inhibition | 25-45 seconds | 1v1, Best of Three

Play: Follow green-light commands while ignoring convincing fake states, moving targets, and rule flips announced before each heat.

Scoring: Correct taps, missed targets, and false taps; highest score with reaction-time tiebreak.

Detection: Touch position/time, rendered target state, device orientation, and integrity signals.

Anti-cheat: Random target paths, app integrity, automation detection, and no predictable cadence.

Showmanship and Highlight: Fakeout failures spring a trap animation around the player's video reaction.

Failure and No Contest: Cancel if frame rate or touch sampling falls below the device-specific floor.

Safety and accessibility: Large-target accessible mode changes scoring cohort; reduced motion removes moving backgrounds.

Analytics and launch gate: False-tap rate, target-size cohort fairness, frame pacing, and rematch appetite.

8.27 Balance Blitz

Motion / control | 30-60 seconds | 1v1, Gauntlet

Play: Tilt the phone to keep a virtual ball inside a shrinking safe zone while symmetric disturbances are announced.

Scoring: Time in zone, edge distance, and recovery after modifiers. Highest normalized control score wins.

Detection: Gyroscope, accelerometer, device orientation, touch state, and sensor integrity.

Anti-cheat: Detect flat-table automation, sensor replay, rooted/jailbroken injection, and impossible zero-noise traces.

Showmanship and Highlight: Opponent tracks are visible as ghost trails; last-second saves become the Highlight.

Failure and No Contest: Run a five-second sensor calibration; No Contest if drift exceeds threshold.

Safety and accessibility: Designed for stationary seated play; never requires walking while looking at the screen.

Analytics and launch gate: Drift, device model variance, modifier recovery, table-cheat flags, and close finishes.

8.28 Step Showdown

Movement / steps | 45-90 seconds | party 1v1, Duos, parties

Play: Accumulate the most valid steps during a short interval while the phone remains secured. Optional rhythm modifiers reward cadence control.

Scoring: Validated steps plus cadence objectives; cap implausible bursts and normalize approved device-placement modes.

Detection: Pedometer, accelerometer, gyroscope, motion classification, device placement, and app foreground state.

Anti-cheat: Reject hand shaking, mechanical vibration, sensor replay, vehicle movement, and impossible cadence.

Showmanship and Highlight: Live step meters race while video shows upper-body reactions; final three seconds use a dramatic sprint meter.

Failure and No Contest: No Contest if device placement changes repeatedly, motion permissions fail, or movement becomes unsafe.

Safety and accessibility: Preflight asks for clear space; audio/haptic cues minimize screen looking. Seated arm-movement alternative is unranked against step mode.

Analytics and launch gate: Step validation precision, shake-cheat rate, placement failures, opt-out rate, and physical-safety reports.

8.29 Distance Dash

GPS / endurance | 60-90 seconds | party 1v1, Duos, event

Play: Cover the greatest verified outdoor distance during the timer. The opponent sees distance and pace only - never a map, route, or nearby status.

Scoring: Fused GPS and motion distance with accuracy weighting; suspicious segments are excluded before comparison.

Detection: Foreground precise location, accelerometer, gyroscope, pedometer, speed, GPS accuracy, route continuity, and screen state.

Anti-cheat: Detect spoofed locations, impossible acceleration, vehicle speed, teleportation, treadmill mismatch, sensor injection, and backgrounding.

Showmanship and Highlight: Abstract progress lanes race without geographic context. A comeback graph and finish reaction create the Highlight.

Failure and No Contest: Cancel when accuracy degrades, vehicle-speed movement appears, the phone is unsafe to handle, or route privacy cannot be guaranteed.

Safety and accessibility: Visible location indicator, screen-lock-friendly audio, and immediate raw-route deletion after the scoring/appeal window.

Analytics and launch gate: Valid-distance yield, spoof flags, accuracy cancellations, ad-hoc route exits, safety stops, and completion by environment.

8.30 Head-Shoulders-Knees-GO!

Body / reaction | 30-50 seconds | 1v1, party elimination

Play: Touch or point to prompted body targets, but react only when the announcer says GO. Fakeouts speed up over time.

Scoring: Correct target, motion onset, and false-start penalties across seven prompts.

Detection: Upper/full-body pose, hand-to-target proximity, cue clock, and motion onset.

Anti-cheat: Random cue ordering, server timing, live motion, and minimum return-to-ready position.

Showmanship and Highlight: False starts receive giant arrows and a buzzer; match point uses a dramatic delayed GO.

Failure and No Contest: Use upper-body-only targets when full framing fails. No Contest if shared accessible target set is too small.

Analytics and launch gate: False-start rate, target recognition, seated-mode fairness, cue comprehension, and party elimination pace.

9. One-hundred-game catalog

The launch pack above supplies games 1-30. The 70 roadmap concepts below complete the initial 100-game product catalog. Each concept still requires prototype playtests and detector validation before activation.

Face and reaction

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Poker Face Pressure	Answer absurd yes/no prompts without changing expression. 30-45s / 1v1	Face baseline + expression change; penalize camera occlusion.	Highlight: Confession-style reaction reveal. Guardrail/accessibility: No sensitive personal prompts; speech-free option. Roadmap: P1
Wink Race	Wink the requested eye and sequence faster than the rival. 25-40s / 1v1	Eye landmarks + chirality; liveness blocks loops.	Highlight: Rapid alternating-eye photo finish. Guardrail/accessibility: Skip for incompatible facial mobility. Roadmap: P1
Eyebrow Olympics	Hit raised, lowered, and alternating brow targets. 30-45s / 1v1	Brow landmarks normalized to baseline.	Highlight: Comic scorecards and close-ups. Guardrail/accessibility:

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			Gesture alternative available. Roadmap: P2
Lip Sync Lock	Silently mouth a prompted phrase for the opponent to decode. 45-75s / 1v1/party	Lip landmarks + constrained phrase set; no raw speech needed.	Highlight: Wrong guesses and reveal montage. Guardrail/accessibility: Curated phrases; caption-first. Roadmap: P2
Side-Eye Showdown	Look toward rapidly changing targets without turning the head. 25-40s / 1v1	Gaze approximation + head pose; device calibration.	Highlight: Suspicious side-eye freeze frame. Guardrail/accessibility: Do not require precise medical-grade gaze. Roadmap: P2
Smile Switch	Switch between neutral and smile exactly on beat. 30-50s / 1v1	Expression transition clock; baseline fairness.	Highlight: Beat-synced expression montage. Guardrail/accessibility: Alternate hand-signal mode. Roadmap: P1
Face Memory	Recreate a three-expression sequence from memory. 40-60s / 1v1	Expression sequence + timing; server-seeded targets.	Highlight: Sequence replay with the first miss. Guardrail/accessibility: Non-appearance-based scoring. Roadmap: P1

Hand and pose

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Finger Twister	Place requested fingers into unusual but safe combinations. 30-50s / 1v1	Hand landmarks; reject offensive gesture classes.	Highlight: Tangled-hand close-up. Guardrail/accessibility: Mobility calibration and skip set. Roadmap: P1
Statue Switch	Jump between two safe poses only on the valid switch cue. 35-55s / 1v1	Pose classification + motion onset.	Highlight: Fakeout switch failures. Guardrail/accessibility: Seated pose set. Roadmap: P1
Shadow Boxer	Hit virtual targets with controlled air punches. 40-60s / 1v1	Wrist trajectory + target collision; no contact.	Highlight: Combo counter and slow-motion finish. Guardrail/accessibility:

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			Clear-space check; no force scoring. Roadmap: P2
Gesture Decoder	Translate a shown symbol into the matching gesture. 30-45s / 1v1	Hand/pose templates; server randomization.	Highlight: Rapid correct-answer snaps. Guardrail/accessibility: Culturally reviewed gesture set. Roadmap: P1
High-Five Sync	Pairs time an air high-five across split screens. 30-45s / Duos	Wrist path and shared impact timestamp.	Highlight: Seamless split-screen high five. Guardrail/accessibility: No physical partner contact required. Roadmap: P1
Pose Ladder	Complete five poses in ascending difficulty. 45-75s / 1v1	Pose confidence + hold duration.	Highlight: Final-pose victory card. Guardrail/accessibility: Space and mobility filtering. Roadmap: P1
Hand Maze	Move a tracked hand through a virtual path without touching edges. 30-60s / 1v1	Hand landmark trajectory + device latency.	Highlight: Last-second maze save. Guardrail/accessibility: Large-path accessible cohort. Roadmap: P2

Scavenger

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Shape Hunt	Find a circle, triangle, square, then a wildcard shape. 45-75s / 1v1	Object boundary/shape detector; reject screen art.	Highlight: Real-world shape montage. Guardrail/accessibility: Safe nearby objects only. Roadmap: P1
Something Tiny	Find the smallest valid object before time expires. 45-60s / 1v1/party	Object scale relative to hand; peer vote only in casual.	Highlight: Tiny-object macro reveal. Guardrail/accessibility: Exclude pills, sharp items, and personal data. Roadmap: P2
Soft vs Hard	Race through alternating material-property targets. 50-80s / 1v1	Texture + object motion; touch confirmation.	Highlight: Split-screen material contrast. Guardrail/accessibility: No

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			hot, sharp, or unknown objects. Roadmap: P2
Brandless Bingo	Complete targets without showing logos or labels. 60-90s / squads	Object classes + logo/text redaction.	Highlight: Creative workarounds. Guardrail/accessibility: Privacy-friendly by design. Roadmap: P2
Number Hunt	Find printed or physical numbers matching a sequence. 45-75s / 1v1	OCR + scene privacy redaction.	Highlight: Rapid number lock-ins. Guardrail/accessibility: Block addresses, IDs, and financial documents. Roadmap: P2
Letter Builder	Arrange safe objects into the target letter shape. 60-90s / 1v1/party	Object segmentation + spatial shape.	Highlight: Before/after construction reveal. Guardrail/accessibility: Small lightweight objects. Roadmap: P2
Mystery Material	Infer and find a material from clues. 45-75s / 1v1	Texture/material detection + clue state.	Highlight: Final clue comeback. Guardrail/accessibility: No chemical or unknown substances. Roadmap: P2

Voice and music

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Fast Rhyme	Alternate valid rhymes until someone stalls or repeats. 40-75s / 1v1	Constrained ASR + rhyme dictionary; repetition check.	Highlight: Unexpected rhyme reaction. Guardrail/accessibility: Locale-specific vocabulary; no slurs. Roadmap: P1
One-Word Song	Sing a melody using only the revealed word. 45-60s / 1v1/party	Voice activity, pitch contour, peer vote in casual.	Highlight: Ridiculous performance montage. Guardrail/accessibility: No vocal-quality ranking. Roadmap: P2
Vocal Copycat	Imitate a safe synthetic sound pattern. 30-50s / 1v1	Spectral similarity + rhythm.	Highlight: Waveform side-by-side. Guardrail/accessibility: No impersonation of real

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			people. Roadmap: P1
Volume Control	Hold voice volume inside a moving target zone. 30-50s / 1v1	Calibrated RMS + noise floor.	Highlight: Meter near-misses. Guardrail/accessibility: Safe volume cap and silent alternative. Roadmap: P2
Voice Character	Match pitch, speed, and rhythm of a fictional voice card. 40-60s / 1v1	Prosody metrics; no identity comparison.	Highlight: Character transformation edit. Guardrail/accessibility: No ethnic or disability caricatures. Roadmap: P2
Category Blitz	Name unique items in a safe category without pausing. 45-75s / 1v1	ASR + semantic category + duplicate check.	Highlight: Stall buzzer and comeback streak. Guardrail/accessibility: Curated age-appropriate categories. Roadmap: P1
Finish the Phrase	Complete original phrase setups with the expected ending. 30-50s / 1v1	Constrained ASR; server-seeded prompts.	Highlight: Confident wrong-answer reveals. Guardrail/accessibility: Original copy; no copyrighted lyrics. Roadmap: P1

Touch and reflex

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Tilt Sprint	Steer through a short generated course faster than the opponent. 30-60s / 1v1	Gyro + collision state; integrity attestation.	Highlight: Ghost-track overtake. Guardrail/accessibility: Stationary seated play. Roadmap: P1
Gyro Goal	Tilt a ball into moving goals in the right order. 40-60s / 1v1	Gyro physics + target sequence.	Highlight: Final-goal buzzer beater. Guardrail/accessibility: Calibration and large-goal mode. Roadmap: P1
Swipe Sequence	Repeat increasingly long directional swipe chains. 30-50s / 1v1	Touch path + sequence timing.	Highlight: Exact wrong swipe highlight. Guardrail/accessibility:

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			Large gesture setting. Roadmap: P1
Perfect Stop	Stop a moving meter as close to the target as possible. 20-35s / tournament	Render/touch timestamp + refresh normalization.	Highlight: Millisecond photo finish. Guardrail/accessibility: No claim beyond measured precision. Roadmap: P1
Timing Tower	Stack moving blocks through precise timed taps. 45-75s / 1v1	Deterministic physics + input telemetry.	Highlight: Tower collapse reaction. Guardrail/accessibility: Reduced-motion presentation. Roadmap: P2
Shake and Settle	Shake to charge, then become perfectly still. 30-45s / 1v1	Accelerometer energy + settle time.	Highlight: Wild shake to instant freeze. Guardrail/accessibility: Wrist strain limit; accessible tap mode. Roadmap: P2
Fakeout Button	Tap the button only when its label and color agree. 25-45s / 1v1	Touch + rendered state; automation detection.	Highlight: Fakeout chain failure. Guardrail/accessibility: Text/icon redundancy. Roadmap: P1

GPS and outdoor

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Sprint to 100	First to a verified 100-meter total wins. 45-120s / party 1v1	GPS-motion fusion; speed/spoof detection.	Highlight: Abstract finish-line comeback. Guardrail/accessibility: Safe-route confirmation; no maps. Roadmap: P1
Walk-Off	Most verified walking distance at a target pace. 2-3m / party 1v1	GPS + pedometer + cadence.	Highlight: Pace-lane overtake. Guardrail/accessibility: Audio cues and screen-off support. Roadmap: P1
Route Roulette	Hit randomized pace zones without revealing a route. 2-4m / party 1v1	Pace state + GPS quality; no geography exchange.	Highlight: Perfect pace-zone streak. Guardrail/accessibility: No

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			directional commands. Roadmap: P2
Elevation Rush	Gain the most verified elevation on a safe route. 3-5m / event	Barometer/GPS elevation + motion.	Highlight: Climb graph without map. Guardrail/accessibility: Opt-in adults first; no stair sprint prompts. Roadmap: P3
Pace Keeper	Stay closest to a revealed target pace. 2-3m / party 1v1	Fused pace + accuracy weighting.	Highlight: Final pace-lock reveal. Guardrail/accessibility: Walking default; running optional. Roadmap: P1
Team Distance Relay	Squad members contribute sequential distance legs. 5-10m / squads	Signed per-leg distance + handoff state.	Highlight: Team progress and final handoff. Guardrail/accessibility: No route sharing; asynchronous fallback. Roadmap: P2
GPS Red Light	Move on green intervals and stop safely on red. 90s / party 1v1	Speed/step state + cue clock.	Highlight: Stop-line photo finish. Guardrail/accessibility: Audio/haptic cues; no screen focus. Roadmap: P2

Creative

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Doodle Dash	Draw a prompted object in ten seconds for model or peer recognition. 30-60s / 1v1/party	Touch strokes + image classifier; peer vote in casual.	Highlight: Bad drawing reveal. Guardrail/accessibility: Safe prompt library. Roadmap: P1
Caption Clash	Write the funniest safe caption for a generated scene. 60-90s / party	Text moderation + peer vote.	Highlight: Caption reveal and reactions. Guardrail/accessibility: No real-person ridicule; links blocked. Roadmap: P2
Prop Character	Turn a safe object into a fictional character performance. 60-90s / party	Object presence + timed performance; peer vote.	Highlight: Character montage. Guardrail/accessibility: No

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			personal or intimate items. Roadmap: P2
Three-Object Story	Tell a short story using three detected objects. 60-90s / party	Object verification + timed speech; peer vote.	Highlight: Plot-twist clip. Guardrail/accessibility: Private information redaction. Roadmap: P2
Paper Sculpture	Build the prompted simple paper shape fastest. 60-90s / 1v1	Shape/object verification.	Highlight: Construction time lapse. Guardrail/accessibility: Paper only; no scissors required. Roadmap: P2
Sound Effects Studio	Perform sound effects for a visual scene. 45-75s / party	Audio presence + peer vote; moderation.	Highlight: Scene plus live Foley edit. Guardrail/accessibility: Safe volume and no frightening prompts. Roadmap: P2
Commercial Pitch	Pitch a ridiculous fictional product in 20 seconds. 60-90s / party	Timer + safe prompt + peer vote.	Highlight: Best sales line clip. Guardrail/accessibility: No financial, body, or adult products. Roadmap: P2

Bluff and deduction

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Two Truths Speedrun	Give two safe claims and one fiction; rival guesses quickly. 60-90s / 1v1/party	Structured input + moderation + vote.	Highlight: Confident wrong guess. Guardrail/accessibility: No sensitive identity or trauma prompts. Roadmap: P2
Fake Object	Show three objects and bluff which one matches the secret property. 60-90s / 1v1	Object presence + hidden role state.	Highlight: Bluff reveal. Guardrail/accessibility: Safe-object list. Roadmap: P2
Guess My Choice	Predict the opponent's answer to a funny forced choice. 30-50s / 1v1	Private simultaneous choice.	Highlight: Matching-reaction montage. Guardrail/accessibility: No sexual, political, or

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			sensitive choices. Roadmap: P1
Bluff Gesture	One player performs the real prompt among decoys. 45-75s / party	Role state + gesture validation.	Highlight: Accusation and reveal. Guardrail/accessibility: Culturally safe gestures. Roadmap: P2
Secret Rule	Infer the hidden rule from accepted and rejected actions. 60-90s / party	Server rule engine + event validation.	Highlight: Final deduction reveal. Guardrail/accessibility: Simple non-sensitive rules. Roadmap: P2
Odd One Out	Identify which player received a different prompt. 60-90s / group	Role distribution + vote.	Highlight: Group accusation montage. Guardrail/accessibility: No appearance-based prompts. Roadmap: P2
Prediction Duel	Predict the opponent's performance outcome before the mini-game. 30-60s / 1v1	Precommit prediction + game result.	Highlight: Called-it reaction. Guardrail/accessibility: No money or transferable stakes. Roadmap: P1

Duo coordination

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Sync Clap	Two teammates clap the same rhythm in sync. 30-50s / 2v2	Cross-client onset alignment.	Highlight: Perfect sync burst. Guardrail/accessibility: Visual/tap alternatives. Roadmap: P1
Mirror Relay	Partners alternate mirrored pose segments. 45-75s / 2v2	Pose sequence + role state.	Highlight: Four-way synchronized finish. Guardrail/accessibility: Seated vocabulary. Roadmap: P1
Split Scavenger	Each partner finds half of a combined target set. 60-90s / 2v2	Object verification + team board.	Highlight: Last-item comeback. Guardrail/accessibility: Safe common objects. Roadmap: P1

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Count Without Colliding	Partners count upward without speaking at the same time. 45-90s / Duos	Voice onset + word recognition.	Highlight: Collision buzzer reactions. Guardrail/accessibility: Tap alternative. Roadmap: P2
Pass the Beat	Partners alternate beats without breaking tempo. 45-75s / 2v2	Cross-client beat clock.	Highlight: Perfect alternating streak. Guardrail/accessibility: Latency-normalized scoring. Roadmap: P1
Duo Pose Lock	Partners form complementary split-screen poses. 45-75s / 2v2	Pose templates across participants.	Highlight: Combined silhouette card. Guardrail/accessibility: No physical contact required. Roadmap: P2
Call-and-Response Race	One partner performs a pattern; the other answers instantly. 45-75s / 2v2	Audio/gesture sequence + role timing.	Highlight: Rapid team exchange. Guardrail/accessibility: Accessible input matching. Roadmap: P1

Group and elimination

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
Last Face Standing	Expression challenges eliminate the first player to break. 3-6m / 4-8	Face change + round state.	Highlight: Multi-reaction elimination grid. Guardrail/accessibility: Safe prompts and no appearance judging. Roadmap: P1
Team Charades	One performer acts; teammates race to choose the answer. 3-8m / 4-8	Safe prompt + timed answer; no model judging.	Highlight: Wild performance reveal. Guardrail/accessibility: Moderated prompt library. Roadmap: P1
Scavenger Relay	Players retrieve one object each in sequence. 4-8m / 4-8	Team board + object validation.	Highlight: Final runner comeback. Guardrail/accessibility: Safe object sets. Roadmap: P1
Crowd Freeze	All move; last person to stop is eliminated. 3-6m / 4-8	Per-player motion + cue clock.	Highlight: Grid-wide slow motion. Guardrail/accessibility:

Game	Competitive hook and format	Detection and anti-cheat	Highlight, guardrail, priority
			Low-impact movement. Roadmap: P1
Captain Says	Team captains issue approved commands against the clock. 4-8m / squads	Template selection + pose validation.	Highlight: Captain bluff and team chaos. Guardrail/accessibility: No free-form unsafe commands. Roadmap: P2
Saboteur Signal	A secret saboteur subtly introduces wrong gestures. 5-10m / 5-8	Role state + gesture chain + vote.	Highlight: Saboteur reveal montage. Guardrail/accessibility: No harassment or targeting mechanics. Roadmap: P2
Chaos Trivia	Fast visual trivia triggers physical answer zones. 3-8m / 4-8	Answer selection + pose zone.	Highlight: Confident wrong-side reactions. Guardrail/accessibility: Age-appropriate original questions. Roadmap: P2

10. Game selection and live-content engine

10.1 Selection rules

- Hard eligibility: age tier, mode, permissions, device sensors, camera framing, accessibility settings, available space, network quality, and current game health.
- Novelty: exclude the previous ten games unless the player chooses a rematch; cap a game family at two appearances in any five-game window.
- Preference: use explicit likes, dislikes, and playlist choices before behavioral inference.
- Fairness: select a game version supported by both devices and by both players' calibrated inputs.
- Content health: remote-disable games or targets when scoring, safety, or detector telemetry degrades.
- Exploration: reserve a controlled portion of eligible rounds for new variants, clearly marked as beta when necessary.

10.2 Game Definition lifecycle

1. Concept and fun hypothesis.
2. Paper or human-refereed prototype.
3. Detector feasibility and privacy review.

4. Internal dogfood with event logging.
5. Moderated teen and young-adult playtest.
6. Limited private-party release.
7. Unranked party beta release.
8. Ranked enablement after fairness gate.
9. Live monitoring, tuning, seasonal remix, or retirement.

10.3 Content cadence

- Two new games or meaningful variants per month after beta stabilization.
- One featured playlist per week with a distinct emotional promise: speed, comedy, scavenger, movement, sound, or squads.
- Eight-week seasons with a theme, known reward track, creator event, and one marquee game.
- Monthly retirement review for low-fun, high-dispute, repetitive, unsafe, or technically unreliable content.
- Community voting and prompt submissions are moderated content inputs, never executable code.

11. Havoc Highlights and virality

Highlights convert the emotional peak of a live match into acquisition. The system should feel like an instant sports replay editor, not a recording archive.

Highlight growth loop: Funny competition -> candidate cut -> every featured participant approves -> vertical export -> deep-linked friend-party invitation -> new competition.

11.1 Candidate generation

- Maintain an encrypted ephemeral round buffer locally.
- Detect candidate beats: close finish, comeback, scoring swing, model-confirmed fail, synchronized reaction, winner pose, and audience reaction.
- Generate three edit options: reaction cut, close-finish cut, and full-round cut where duration allows.
- Apply game title, countdown, animated score, winner stamp, captions, and restrained Havoc branding.
- Never upload or retain a public clip until all featured players approve.

11.2 Consent and identity

- Every featured party member reviews the exact clip before public export.
- Handles, age bands, background details, and social usernames are hidden by default.
- A player may approve gameplay use while declining identity display.

- If any featured participant declines, the multi-person public version is deleted; participant-approved crops may be offered when technically and legally valid.
- Guardian controls may disable public export for ages 13-15 private-party accounts.

11.3 Distribution

- Vertical 9:16 export for TikTok, Reels, Shorts, Snapchat, and Messages.
- Deep link to the exact game, creator challenge, or private party invitation.
- Challenge templates let viewers record their own attempt after installation.
- Track approval, export destination, referral attribution, install, onboarding completion, and first challenge.
- Do not make Highlights an infinite passive feed during beta; they support play rather than replace it.

12. Party GPS and physical competition

Product rule. GPS games launch only inside invite-only parties. Location is a scoring input - never a friend-discovery, map-sharing, or proximity surface. Public GPS competition belongs to the separately gated future random-match phase.

12.1 Privacy and data flow

- Request foreground precise location only when a player explicitly enters a GPS game.
- Process route geometry on-device; transmit signed distance, time, pace, accuracy, and anti-cheat summaries.
- Never show coordinates, route lines, neighborhood, distance from opponent, or 'nearby' status.
- Delete raw route data after the scoring and appeal window; do not reuse it for advertising or discovery.
- Display an unmistakable active-location indicator and emergency stop.

12.2 Safety and fair play

- Preflight: clear route, safe surroundings, secure phone placement, no driving, no prohibited environment, and audio/haptic cue readiness.
- Detect vehicle speeds, impossible acceleration, teleportation, spoofing frameworks, sensor replay, treadmill mismatch, and backgrounding.
- Cancel rather than score when the system detects risk or cannot verify the movement.
- Do not issue turn-by-turn or direction prompts. Havoc compares effort, not geography.
- Use pace and step modes to create competitive variety without forcing maximum sprinting.

Privacy basis. The product follows the principle that child geolocation should default off and be obvious when active. [ICO Children's Code geolocation guidance](#) informs the default-off and session-scoped design.

13. ML scoring, anti-cheat, and game quality

13.1 Shared perception primitives

Primitive	Capabilities	Example consumers
Face	Presence, landmarks, blink, expression, head pose, capture quality, liveness	Stare, Don't Laugh, Emoji, Meme
Hand	Landmarks, finger states, chirality, gesture, trajectory	RPS, signals, hand maze
Body	2D/3D pose, velocity, joint angles, motion energy, segmentation	Pose, Freeze, dance, Simon
Object	Curated detection, segmentation, OCR, color, texture, material	Scavenger family
Audio	ASR, onset, clap, pitch, rhythm, event class, replay fingerprints	Voice and music family
Device	Touch, render timing, gyro, accelerometer, pedometer, integrity	Reflex and balance
Location	GPS-motion fusion, pace, distance, accuracy, spoof detection	Public outdoor family

13.2 Scoring architecture

1. On-device models produce time-stamped observations and confidence, not final trust by themselves.
2. The game client converts observations into versioned candidate events.
3. The server-authoritative match clock aligns both clients and applies the Game Definition.
4. Anti-cheat and network-quality services annotate events with integrity and timing confidence.
5. The scoring service returns win, loss, draw, or No Contest plus an explanation code.
6. Audited samples and appeals feed model evaluation and content tuning, not automatic punishment without policy.

13.3 Launch quality gates

Metric	Beta gate	Release behavior
Audited false winner	< 1%	No ranked enablement above threshold
Inconclusive result	< 3%	Tune calibration or remove unreliable targets
Crash-free sessions	> 99.8%	Block release on regression
Match completion	> 85%	Segment exits by game, network, ad, and safety

Metric	Beta gate	Release behavior
Media join p95	< 4 seconds	Regional and device monitoring
Domestic media latency p95	< 400 ms	Use network quality and adaptive bitrate
Detector frame budget	Target $\leq$ 50 ms on supported devices	Dynamic sampling and capability tiers
High-severity moderation action	Target < 1 second from confident trigger	Blur, mute, pause, or terminate

13.4 Evaluation and bias

- Test across supported device classes, frame rates, lighting, skin tones, clothing, backgrounds, accents, languages, mobility, assistive modes, and network conditions.
- Evaluate false positive, false negative, false winner, and No Contest rates separately.
- Maintain a model card and data statement for every custom detector.
- Do not train on teen production media without explicit, separately reviewed consent and legal basis.
- Use consenting staged data, synthetic augmentation, and privacy-preserving telemetry wherever possible.
- Ship per-model rollback and per-game confidence-threshold feature flags.

Feasibility references. [Apple Vision](#) provides face, text, object, and pose analysis primitives, while [MediaPipe Pose Landmarker](#) supports cross-platform pose inference patterns. These are starting points, not proof that any Havoc scoring model is production-ready.

14. Parties, tournaments, rivals, rankings, and progression

14.1 Party rooms

- Support 2-8 invited active players with host-selected playlists; Party Mode is the launch product, not a secondary feature.
- Create an expiring deep link and short code for every room; teams may be fixed, randomized, captained, or reshuffled after each block.
- Late joiners enter between rounds; they never disrupt a live scoring state.
- Host controls: remove, mute, reorder playlist, select team format, enable accessible modes, and end the party.
- Safety controls remain individual and cannot be disabled by the host.

14.2 Rankings

- Family ratings prevent a specialist in Reaction Flash from dominating a first-time scavenger player.
- Launch rankings emphasize friend head-to-head and party-season records; public rank remains disabled until stranger matchmaking is separately approved.
- Seasonal rank soft-resets while lifetime records remain.
- Friend and party leaderboards never expose exact minor identity, school, location, or activity schedule; global boards remain out of launch scope.
- No rank loss for system-caused No Contest, verified interruption, or safety termination.

14.3 Rivalries and sportsmanship

- Mutual rivals receive head-to-head record, favorite shared games, and rematch prompts.
- Sportsmanship signals include completed losses, low report history, valid play, and positive post-match reactions.
- Do not turn reports into public reputation points or enable retaliatory downvoting.
- Repeated rage quitting creates party cool-downs or host rejoin limits only after distinguishing technical failures.

15. Advertising, subscriptions, currency, and sponsorships

15.1 Advertising policy

- Create one interstitial opportunity after every two completed standard games.
- Do not split a Best of Three, decisive Gauntlet finale, active tournament round, report, appeal, or Highlight consent flow.
- Rewarded ads grant known cosmetic value, temporary party themes, or Highlight styles - never score, rank, or matchmaking benefits.
- For minors, use contextual age-appropriate advertising and compliant SDK configurations; do not use precise location for ads.
- When inventory fails suitability checks, show a house ad for a new game, season, creator pack, or party feature.
- Remote-configure frequency so the team can respond if the two-game cadence damages fun, retention, or brand safety.

15.2 Products and planning prices

Product	Planning price	Value
Havoc No Ads	\$1.99 monthly / \$14.99 yearly	Removes interstitial and banner inventory; house safety notices remain.
Havoc+	\$4.99 monthly / \$34.99 yearly	No ads, cosmetic drop, premium Highlight styles, stats, party themes, reactions, game voting.
Season Pass	\$4.99 per eight-week season	Known free and premium reward tracks; no random rewards.
Havoc Coins	Direct purchase packs	Known cosmetics, entrances, frames, reactions, result effects, and party themes.
Creator packs	One-time or sponsored	Curated games, prompts, Highlight frames, and event access.
Brand challenges	Sponsorship	Age-appropriate labeled mini-games or tournaments with no gameplay advantage.

15.3 Monetization guardrails

- Do not sell wins, easier opponents, model confidence, safety access, faster reports, teen visibility, or public location.
- No loot boxes, cash-out, transfers, secondary market, or wagering.
- Show exact price and item before purchase; parental purchase controls follow platform requirements.
- Track ad-driven exits, session compression, fun-score changes, conversion, refund, chargeback, and support burden.
- Treat pricing and conversion targets as hypotheses to test, not guaranteed forecasts.

Comparable pattern. Yubo combines subscriptions with virtual currency and visibility products. Havoc borrows the mixed model but rejects paid matchmaking advantage. See [Yubo subscriptions](#) and [YuBucks](#).

16. Trust, moderation, privacy, and account integrity

Trust systems should be felt as clean matches, real opponents, fast recovery, and correct outcomes. The main interface remains playful; the operating system behind it is rigorous.

16.1 Layered controls

- Age assurance, liveness, device reputation, ban-evasion detection, and risk-based re-verification.
- Pre-reveal video moderation while opponent media remains blurred.
- Real-time nudity/sexual content, grooming language, hate, threat, self-harm, violence, scam, and spam signals.

- Immediate local controls: leave, report, block, mute, and camera off.
- Automated actions: warn, blur, mute, pause, terminate, restrict party access or a future public queue, or hold for review.
- Human moderation queue with high-severity prioritization, evidence controls, appeals, and audit trail.
- Published community standards, child-safety point of contact, and required regional reporting procedures.

16.2 Ephemeral evidence

- Keep a locally encrypted rolling buffer sized for recent context, not a routine server recording.
- Upload only after a user report, a high-confidence severe trigger, or all-featured-participant Highlight consent.
- Apply role-based access, immutable audit logging, encryption, retention class, deletion job, and legal-hold process.
- Do not use safety evidence for advertising or casual product analytics.

16.3 Youth well-being

- Default quiet hours and non-urgent notification limits for minors.
- No punitive streak loss, endless autoplay, manipulative privacy prompts, or public popularity counters for minors.
- Break suggestion after sustained play without blocking a safe exit.
- North-star and experiment reviews include safety and well-being guardrails, not only time spent.

Youth design basis. [HHS youth social-media advisory](#), [APA adolescent social-media guidance](#), and [EU minor-protection guidance](#) support high-engagement design without coercive defaults.

Privacy basis. [FTC COPPA FAQ](#) and [ICO guidance on nudge techniques](#) inform the guardian-controlled and high-privacy treatment for younger users.

17. Architecture, performance, security, and reliability

Havoc reference architecture

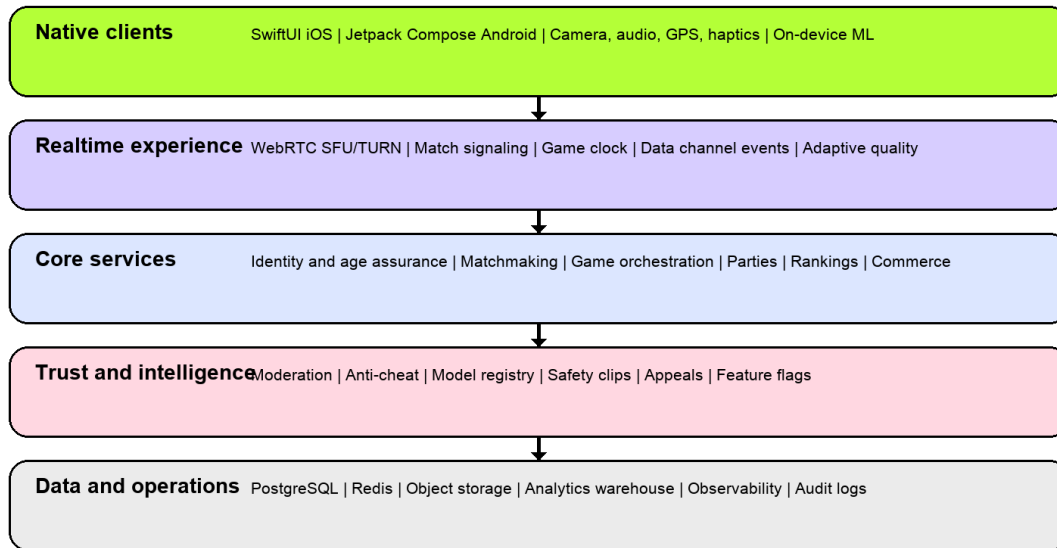


Figure 2. Reference architecture separates media, party orchestration, trust, and durable data.

17.1 Client recommendation

- Native SwiftUI iOS and Jetpack Compose Android clients for camera, media, haptics, sensors, permissions, device integrity, and on-device ML.
- Share protocol definitions, game manifests, analytics names, and model assets where practical; do not force performance-critical camera code through an unsuitable abstraction.
- Capability-tier devices and dynamically reduce video resolution or detector sampling before sacrificing game-clock integrity.

17.2 Realtime media

- Use WebRTC with a managed multi-region SFU and TURN network during beta.
- Use trickle ICE, regional routing, adaptive bitrate, simulcast where supported, network-quality telemetry, and audio-priority degradation.
- Use a reliable data channel or service channel for game events, with server-authoritative timing and idempotent state transitions.
- For group rooms, subscribe only to visible/needed media qualities and preserve safety processing requirements.

Realtime references. [WebRTC peer connections](#) documents ICE, STUN, TURN, and trickle ICE; [LiveKit's SFU architecture](#) illustrates managed adaptive forwarding suitable for a beta reference architecture.

17.3 Core services

- Identity and age assurance; friend graph; invite and party routing; realtime session coordinator; game definition and scoring; tournament; ranking; Highlights; moderation; commerce; notifications; analytics; model registry; feature flags; and a dormant future-matchmaking boundary.
- PostgreSQL for durable relational state, Redis for invite/session coordination, encrypted object storage for approved clips/evidence, and a warehouse for minimized event analytics.
- All public APIs use short-lived tokens, scoped claims, rate limits, replay protection, input validation, and least-privilege service identities.

17.4 Security

- Threat model account takeover, bot farms, malicious media, replayed cameras, sensor injection, GPS spoofing, payment abuse, moderator abuse, scraping, and data exfiltration.
- Use App Attest and Play Integrity signals, TLS, encryption at rest, secrets rotation, signed upload URLs, audit logs, dependency scanning, and isolated moderation access.
- Independent mobile/API penetration test before the friends-first beta; any future stranger-matching beta requires a new dedicated security and abuse review.
- No known critical vulnerabilities at release; documented incident severity, owner, containment, notification, recovery, and postmortem process.
- Backups and disaster recovery are tested, not merely configured.

18. Analytics and fun-testing framework

18.1 North star and guardrails

Area	Metric family	Use
North star	Completed friend-party games per weekly active player	Combines play, completion, and trust.
Fun	Post-round rating, rematch/next rate, replay value	Game-specific and cohort-specific.
Retention	D1, D7, D30; games/session; session return interval	Do not optimize time alone.
Virality	party create, invite send, invite open, join, room start, Highlight share, referral install	Measure full playable loop.
Fairness	False winner, No Contest, dispute, device/identity gap	Launch and rollback gate.

Area	Metric family	Use
Safety	Exposure, report, block, intervention, review time, recurrence	Experiment guardrail.
Revenue	Ad impressions, ad exits, ARPDAU, payer conversion, churn	Segment minors and adults appropriately.

18.2 Fun scorecard

- Target post-round fun rating: at least 4.3/5 before broad release.
- At least 70% choose rematch or another game.
- At least 85% understand the rule without extra explanation.
- Fewer than 5% cite repetition or confusion as the session-ending reason.
- Track close-match rate, funny-failure/Highlight rate, rage quit, dispute, and replay value after 5, 20, and 50 plays.
- Remove or redesign games based on observed boredom even when the concept tested well internally.

18.3 Event taxonomy

Domain	Canonical events
Onboarding	onboarding_started, age_assurance_completed, calibration_completed, permission_result
Match	party_created, invite_sent, party_joined, lobby_ready, preflight_result, game_revealed, round_started, round_completed
Score	observation_emitted, verdict_issued, no_contest, dispute_started, appeal_resolved
Social	friend_requested, rivalry_created, party_joined, rematch_selected, social_reveal
Highlight	candidate_created, approval_result, export_started, share_completed, deep_link_opened
Safety	report_created, block_created, automated_action, moderator_decision, reverify_required
Revenue	ad_opportunity, ad_impression, rewarded_complete, offer_viewed, purchase, renewal, refund
Quality	rtc_join, network_quality, reconnect, detector_latency, crash, battery_thermal_state

18.4 Experiment policy

- Every experiment declares a player benefit hypothesis, primary metric, safety/fairness guardrails, sample, stop condition, and owner.
- Do not experiment on age separation, report access, emergency exit, evidence access, or core privacy defaults.
- The ad-every-two-games policy is remotely configurable and must be evaluated against fun rating, rematch rate, session exit, and subscription conversion.
- Teen experiments receive additional privacy and well-being review.

19. Launch strategy and creator marketing

19.1 Friend-group activation strategy

- Seed friend-group cohorts through schools, teams, clubs, camps, and creator communities; optimize for one host bringing three or more real friends into a playable room.
- Run scheduled Havoc Party Nights with featured playlists, creator hosts, and group rewards that give friends a reason to arrive together.
- Make the invite link the primary growth surface: challenge preview, one-tap join, lightweight onboarding, and direct landing in the intended room.
- Use region for latency and launch operations only; Party Mode never suggests nearby people or strangers to invite.

19.2 Creator launch

- Recruit reaction, comedy, sports, dance, and challenge creators whose audiences fit approved age categories.
- Provide Creator Challenge packs, tournament hosting tools, and branded Highlight frames.
- For minor creators, require appropriate guardian, contract, disclosure, and production safeguards.
- Market real gameplay only; never advertise a fake challenge or a level that users cannot play.
- Use the shared challenge deep link as the CTA, not a generic app-store landing page whenever possible.

19.3 Product-led referral

- Party codes and deep links open the intended room and featured playlist after onboarding.
- New users can see the challenge preview before account creation without exposing other players.
- Referrals grant known cosmetic rewards after the referred player completes a verified game.
- Prevent spam through invite limits, device/account risk, and no access to a user's contacts without explicit permission.

20. Roadmap, staffing, and launch gates

Phase	Planning duration	Build	Exit gate
0. Product and safety definition	4-6 weeks	Prototype top game families; age, policy, threat, and data design; vendor evaluation.	Fun prototypes and safety architecture approved.

Phase	Planning duration	Build	Exit gate
1. Internal playable	8-10 weeks	Core RTC, match state, Practice Lab, six games, scoring telemetry, reports.	Reliable internal sessions and detector feasibility.
2. Friends-only alpha	6-8 weeks	Private parties, 15 games, Highlights prototype, moderation console.	Fun score and stability gates in controlled cohorts.
3. Friends-first closed beta	6-10 weeks	Invite-only iOS and Android parties, 30 games progressively enabled, Highlights, ads, subscriptions, and party GPS.	Invite-to-join conversion, party start success, false-winner rate, incident response, and penetration test.
4. Friends-first app-store launch	6-8 weeks	App-store release, creator party campaigns, reliable invite and deep-link onboarding, guardian controls, and scaled live operations.	Party creation, join, fun, retention, referral, and youth-protection gates signed off.
5. Random-match R&D	8-12 weeks	Policy review, separate age-assurance design, adult-only prototypes, liquidity tests, moderation modeling, and adversarial safety validation.	Independent approval before any stranger-matching beta.
6. Controlled random-match expansion	Ongoing	Friends-first Party Mode remains primary; optional age-separated public matching rolls out only after every separate gate passes.	Policy clearance, safety, liquidity, trust, unit economics, and moderation capacity.

20.1 Core staffing

- Product leadership and live-ops producer.
- Native iOS and Android engineering.
- Backend, realtime media, and infrastructure engineering.
- ML/computer vision, audio ML, data engineering, and model evaluation.
- Game/interaction design, product design, motion, sound, and content writing.
- QA automation, device lab, network testing, accessibility, and adversarial anti-cheat testing.
- Trust and safety product, moderation operations, child-safety expertise, privacy/legal, and security.
- Growth, creator partnerships, lifecycle, advertising operations, commerce, and customer support.

20.2 Launch gates

- Thirty games enabled only when individually passing fun, scoring, accessibility, and safety gates; progressive activation is acceptable inside the controlled beta.
- Party creation, invite delivery, deep-link opening, onboarding, join, reconnect, and room-start flows meet reliability and conversion gates.

- Moderation coverage, severe-incident playbook, reporting obligations, and appeal workflow are staffed.
- Payment, ad, privacy, age, and app-store reviews are complete.
- Independent security test has no unresolved critical issue.
- Rollback, kill switch, status communication, and incident on-call have been exercised.

21. Proposed interfaces and state contracts

21.1 GameDefinition

A versioned manifest consumed by clients, party orchestration, scoring, content operations, analytics, moderation, and the separately gated future matchmaking service.

- Identity: id, displayName, version, family, rolloutState, owner.
- Eligibility: ageBands, regions, modes, playerCounts, deviceCapabilities, permissions, accessibilityTags.
- Presentation: oneLineRule, demoAsset, countdown, durationSec, audioTheme, resultTheme, highlightRules.
- Detection: detectorPipelines, modelVersions, samplingPolicy, calibration, confidenceThresholds.
- Scoring: event schema, point rule, tie rule, simultaneity window, latency policy, No Contest conditions.
- Integrity: antiCheatRules, attestation requirement, plausibility bounds, audit sample rate.
- Safety: challenge class, prohibited contexts, target allowlist, emergency behavior, evidence class.
- Live ops: playlist tags, novelty weight, experiment flags, remote kill switch, minimum client version.

21.2 Match lifecycle

PARTY_LOBBY -> PREFLIGHT -> REVEAL -> COUNTDOWN -> LIVE -> VERIFY -> RESULT
-> HIGHLIGHT / REVENGE_VOTE / NEXT

- Terminal or exceptional states: CANCELLED, DISCONNECTED, SAFETY_TERMINATED, NO_CONTEST, FORFEIT, and EXPIRED.
- All transitions are server authorized, idempotent, time-stamped, and observable.
- Clients render optimistic animation only when it cannot change the authoritative result.

21.3 ScoringVerdict

- matchId, roundId, gameId, gameVersion, verdictId.
- winnerParticipantId or null, resultType, scoreByParticipant, confidence.
- detectorVersions, clockOffsets, latencyAdjustment, integrityFlags, explanationCode.

- evidenceReferences where policy permits, auditState, appealEligibility, issuedAt server timestamp.

21.4 HighlightCandidate

- match and round reference, candidateType, local start/end offsets, featuredParticipants.
- reactionScore, competitiveImportance, renderTemplate, caption tracks.
- per-player approval, identity redaction, background redaction, social-handle display.
- export format, destination intent, expiration, deletion state, referral deep link.

21.5 Additional contracts

Contract	Required responsibility
SafetyEvent	Category, severity, source, confidence, automated action, evidence class, review and escalation.
AdOpportunity	Placement, completed-game count, set state, age class, suitability, suppression reason, result.
PartyState	Members, roles, teams, host controls, playlist, active match, spectator capabilities.
TournamentBracket	Seed policy, rounds, participants, game sets, result source, advancement, interruption.
RivalryRecord	Mutual status, head-to-head, shared family records, rematch eligibility, block state.
ModelCard	Purpose, data, version, metrics, limitations, thresholds, fairness slices, rollback.

22. Major risks and mitigations

Risk	Failure	Mitigation	Priority
App-store classification	Friends-first product is marketed or designed like random chat, or the future mode launches before approval.	Party-first onboarding and navigation, invite-only launch, no stranger surfaces, and a separately gated future-mode review.	P0
Teen exploitation	Grooming or sexual content reaches minors.	Age separation, pre-reveal moderation, real-time actions, staffing, evidence, reports, kill switches.	P0
Scoring distrust	Players see obvious false wins or detector bias.	Per-game gates, confidence, No Contest, audits, appeals, model rollback.	P0
GPS harm/privacy	Unsafe movement or route exposure.	No maps or proximity, foreground opt-in, sensor fusion, safe-route confirmation, emergency stop, deletion.	P0
Content fatigue	Games become repetitive despite catalog size.	No-repeat rule, family balancing, two monthly drops, retire low-fun games.	P0

Risk	Failure	Mitigation	Priorty
Ad fatigue	Ad after every two games kills momentum.	Set-aware placement, remote frequency control, cheap No Ads, house ads, fun guardrail experiment.	P0
Party activation	Hosts fail to bring enough friends into a room or invited users abandon onboarding.	Challenge previews, one-tap deep links, guest-fast onboarding, reminders, warmups, and small-party-compatible games.	P0
Cost	RTC, moderation, inference, and evidence exceed revenue.	Managed beta vendors, sampling, on-device ML, unit-cost dashboards, staged scale.	P1
Name conflict	Havoc cannot be cleared or protected.	Formal trademark and app-store search before brand investment; maintain rename contingency.	P1
Creator misuse	Marketing encourages unsafe or fake challenges.	Contract controls, approved packs, disclosure, real-gameplay policy, rapid takedown.	P1

23. Sources and glossary

23.1 Research sources

- [Apple App Review Guidelines](#) - UGC moderation and random/anonymous chat distribution risk.
- [Apple random-chat guideline update](#) - Clarification that random or anonymous chat is subject to UGC rules.
- [Google Play Child Safety Standards](#) - Requirements for random chat, social, and dating apps.
- [Google Play UGC policy](#) - Reporting, blocking, moderation, and objectionable-content requirements.
- [Google Play Families policy](#) - Child-directed social, sensor, data, and advertising constraints.
- [FTC COPPA FAQ](#) - US child privacy and parental-consent guidance.
- [eSafety statement on Omegle](#) - Operational and moderation warning from Omegle's shutdown.
- [eSafety Safety by Design](#) - Safety as a system and lifecycle responsibility.
- [HHS Social Media and Youth Mental Health](#) - Youth well-being and platform responsibilities.
- [APA Health Advisory](#) - Developmentally appropriate social product design.
- [ICO geolocation guidance](#) - Default-off child geolocation and visible tracking.
- [ICO nudge guidance](#) - Avoid privacy-reducing nudges for children.
- [EU protection-of-minors guidance](#) - Problematic engagement, recommender, and minor protections.
- [Yubo age verification](#) - Age estimation, ID fallback, and behavioral monitoring reference.
- [Yubo subscriptions](#) - Subscription packaging reference.
- [Yubo YuBucks](#) - Virtual currency and power-up reference.

- [Hago Google Play listing](#) - Social games, voice rooms, ads, purchases, and scale reference.
- [Houseparty profile](#) - Spontaneous presence and friend-group adoption.
- [WebRTC peer connections](#) - ICE, STUN, TURN, signaling, and trickle ICE.
- [LiveKit SFU](#) - Managed SFU and adaptive track architecture.
- [Apple Vision framework](#) - On-device face, pose, text, and object analysis.
- [MediaPipe Pose Landmarker](#) - Cross-platform pose inference reference.
- [USPTO trademark search](#) - Preliminary US mark search; not a substitute for legal clearance.

23.2 Glossary

Term	Definition
Game Definition	Versioned configuration describing eligibility, presentation, detection, scoring, integrity, safety, and rollout.
Game family	A skill and detector category used for rating, selection, and accessibility.
No Contest	A result where the system declines to name a winner because fairness or evidence is insufficient.
Havoc Highlight	Participant-approved short replay generated from an ephemeral round buffer.
Public pool	Post-launch, separately gated age-separated matchmaking among users who are not already friends.
Private party	Invite-controlled room that still enforces age, safety, and account policy.
SFU	Selective Forwarding Unit used to route realtime media efficiently.
TURN	Relay service used when a direct WebRTC network path is unavailable.
Liveness	Evidence that camera or sensor input represents a present live participant rather than replayed media.
Fun scorecard	Per-game quantitative and qualitative gate for comprehension, excitement, replay value, and fairness.

23.3 Final product statement

Havoc is the game a friend sends when the group chat needs something to do. One invite opens a live party, the mini-games create the competition and chaos, and every win, loss, comeback, failure, and reaction gives the group a reason to play again and invite someone else.